ADVANCED

Stage 6

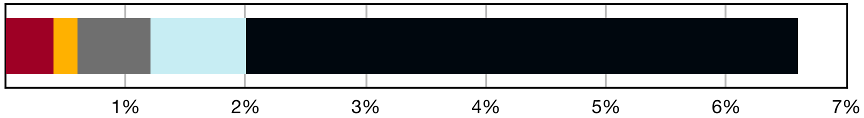
Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Other Chart Types (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 79.5 minutes *(based on allocation of 1.5 minutes per mark)*

S2 Interpretation and Bivariate Data



- Bar Charts and Histograms
- Other Chart Types
- Summary Statistics - Box Plots
- Summary Statistics - No Graph
- Bivariate Data Analysis

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- *S2 Interpretation and Bivariate Data* has contributed an average of 6.6% per new syllabus Advanced exam since 2020.
- We have split the area into six sub-topics for analysis purposes which are: *1-Classifying Data (0%), 2-Bar Charts and Histograms (0.4%), 3-Other Chart Types (0.2%), 4-Summary Statistics - Box Plots (0.6%), 5-Summary Statistics - No Graph (0.8%) and 6-Bivariate Data Analysis (4.6%).*
- This analysis looks at *Other Chart Types.*

HSC ANALYSIS - What to expect and common pitfalls

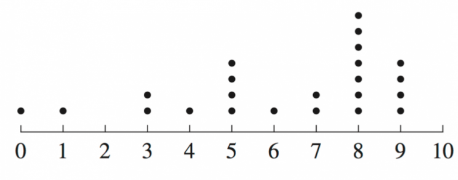
- *Pareto Charts* were examined most recently in new syllabus Advanced exams in 2022 with a 3-mark question that involved frequency tables.
- *Pareto Charts* represents important common content between the Advanced and Std2 courses and possible regular testing in both exams.
- *Stem & Leaf plots* require revision attention and more particularly, *Double Stem and Leaf* plots, which have caused problems in past Std2 exams.
- We believe it is unlikely that examiners will test *Area charts* within the new syllabus, but the generality of the syllabus wording means best practice is briefly covering this graph type.
- Other charts tested that students have answered well include pie charts, segregated bar charts, step graphs and dot plots.

Questions

1. Statistics, STD2 S1 2010 HSC 1 MC

The results of a survey are displayed in the dot plot.

What is the range of this data?



A. 7

B. 8

C. 9

D. 10

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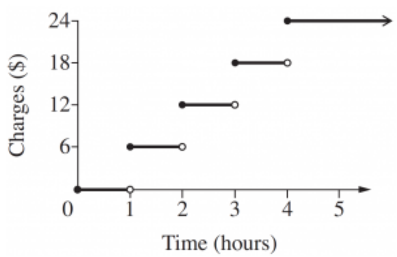
2. Statistics, STD2 S1 2009 HSC 2 MC

The step graph shows the charges for a carpark.



Maria enters the carpark at 10:10 am and exits at 1:30 pm.

How much will she pay in charges?

A. $6

B. $12

C. $18

D. $24

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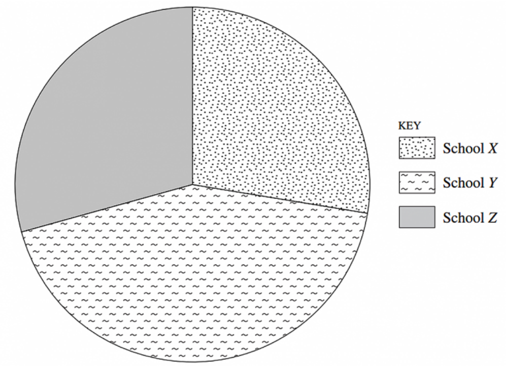
3. Statistics, STD2 S1 2004 HSC 8 MC

This sector graph shows the distribution of 116 prizes won by three schools: X, Y and Z.



How many prizes were won by School X?

A. 26

B. 32

C. 81

D. 99

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4. Statistics, 2ADV S2 SM-Bank 5 MC

The stem plot below shows the height, in centimetres, of 20 players in a junior football team.

key: 14|2 = 142 cm n = 20

14	2	2	4	7	8	8	9	
15	0	0	1	2	5	5	6	8
16	0	1	1	2				
17	9							

A player with a height of 179 cm is considered an outlier because 179 cm is greater than

A. 162 cm

B. 169 cm

C. 173 cm

D. 175.5 cm

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5. Statistics, STD2 S1 2012 HSC 1 MC

A set of 15 scores is displayed in a stem-and-leaf plot.

5	3	4		
6	2	6	7	
7	7	7	8	9
8	2	4		
9	1	3	5	7

What is the median of these scores?

A. 7

B. 8

C. 77

D. 78

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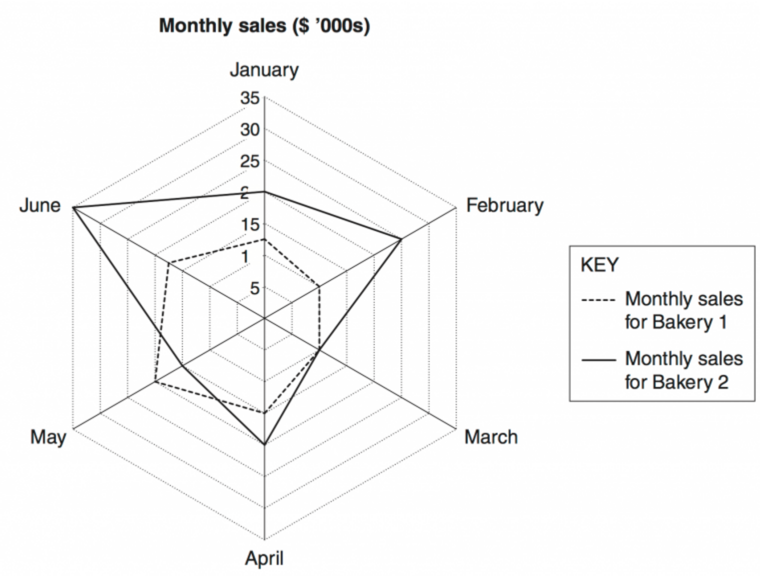
6. Statistics, STD2 S1 2012 HSC 7 MC

The Pi Company has two bakeries. The radar chart displays the monthly sales for the bakeries.



What was the difference in sales in June between the two bakeries?

A. $7.50

B. $17.50

C. $7500

D. $17 500

7. Statistics, STD2 S1 2006 HSC 4 MC

A set of scores is displayed in a stem-and-leaf plot.

1	2 2 3
2	5 8
3	8 9
4	1 3 9

What is the median of this set of scores?

A. 28

B. 30

C. 33

D. 47

8. Statistics, STD2 S1 2008 HSC 3 MC

The stem-and-leaf plot represents the daily sales of soft drink from a vending machine.

If the range of sales is 43, what is the value of N ?

2	N 5 5
3	4 7 7 9
4	0 5 8
5	2
6	0 7

A. 4

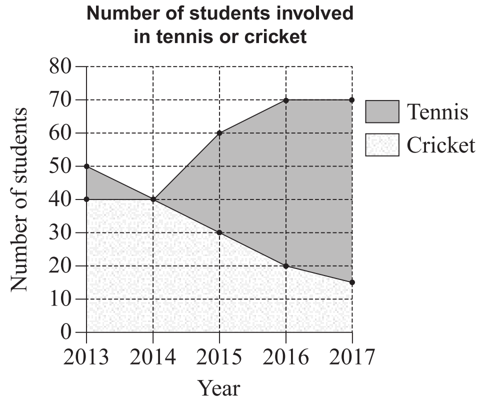
B. 5

C. 24

D. 25

9. Statistics, STD2 S1 2018 HSC 17 MC

The area chart shows the number of students involved in tennis or cricket at a school over a number of years.



In which year was the number of students involved in tennis equal to the number of students involved in cricket?

A. 2013

B. 2014

C. 2015

D. 2016

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10. Statistics, STD2 S1 2010 HSC 16 MC

This back-to-back stem-and-leaf plot displays the test results for a class of 26 students.

Boys		Girls
1	2	1 2 4
3	3	0 2 3 5
9 7 4	4	4 4 5 9 9
6 4 2 2	5	3
3 0	6	1 9

What is the median test result for the class?

A. 44

B. 46

C. 48

D. 49

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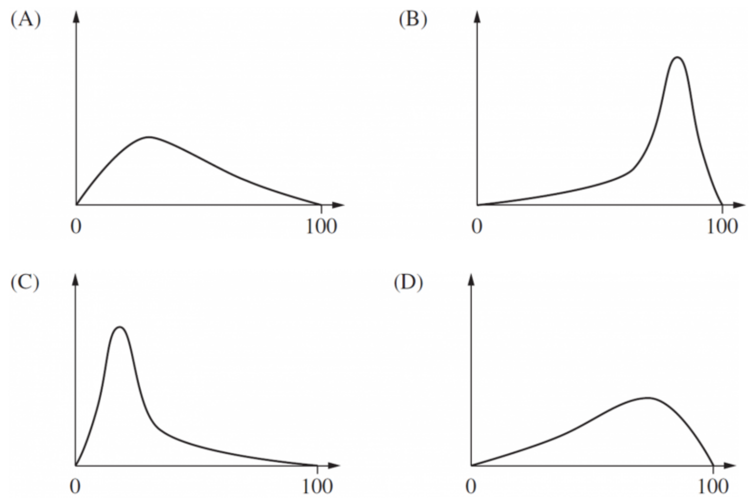
11. Statistics, STD2 S1 2006 HSC 8 MC

Which of these graphs best represents positively skewed data with the smaller standard deviation?



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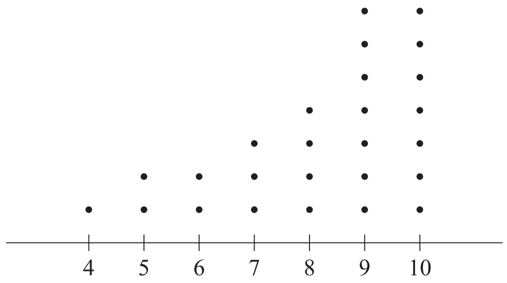
12. Statistics, STD2 S1 2018 HSC 6 MC

A set of data is displayed in this dot plot.



Which of the following best describes this set of data?

A. Symmetrical

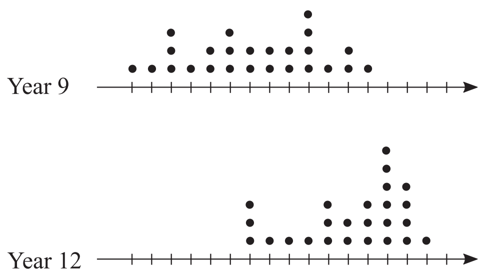
B. Positively skewed

C. Negatively skewed

D. Normally distributed

13. Statistics, STD2 S1 SM-Bank 2 MC

The dot plots show the height of students in Year 9 and Year 12 in a school. They are drawn on the same scale.



Which statement about the change in heights when comparing Y9 to Y12 is correct?

A. The mean increased and the standard deviation decreased.

B. The mean decreased and the standard deviation decreased.

C. The mean increased and the standard deviation increased.

D. The mean decreased and the standard deviation increased.

14. Statistics, STD2 S1 2015 HSC 19 MC

The table shows the life expectancy (expected remaining years of life) for females at selected ages in the given periods of time.

Life expectancy for females

Period of time	*Remaining years of life*			
	at age 0	at age 25	at age 45	at age 65
1965–1967	74.2	51.2	32.3	15.7
1975–1977	76.6	53.1	34.0	17.1
1985–1987	79.2	55.4	36.1	18.6
1995–1997	81.3	57.1	37.7	19.8

In 1975, a 45-year-old female used the information in the table to calculate the age to which she was expected to live. Twenty years later she recalculated the age to which she was expected to live.

What is the difference between the two ages she calculated?

A. 2.7 years

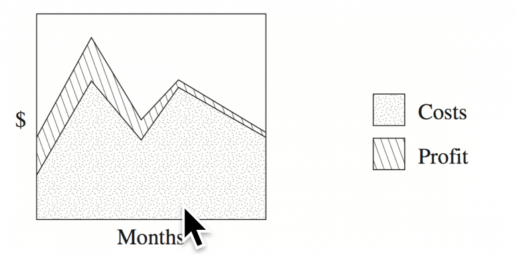
B. 3.1 years

C. 3.7 years

D. 5.8 years

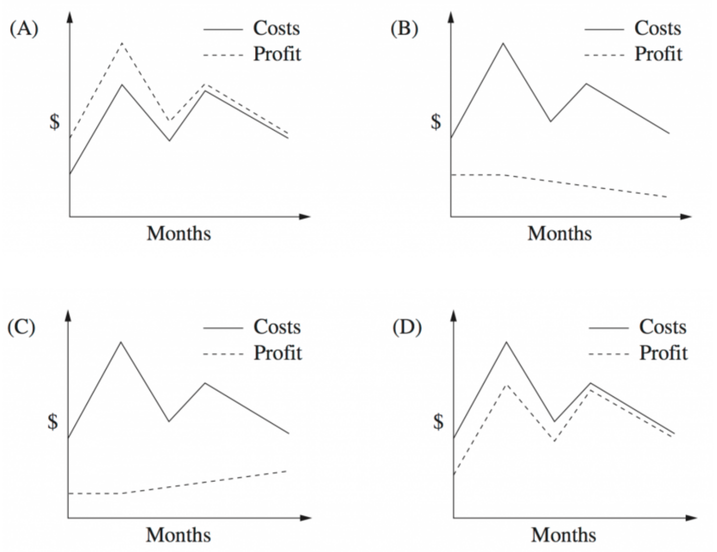
15. Statistics, STD2 S1 2010 HSC 21 MC

The area graph shows the cost and profits for a business over a period of time.



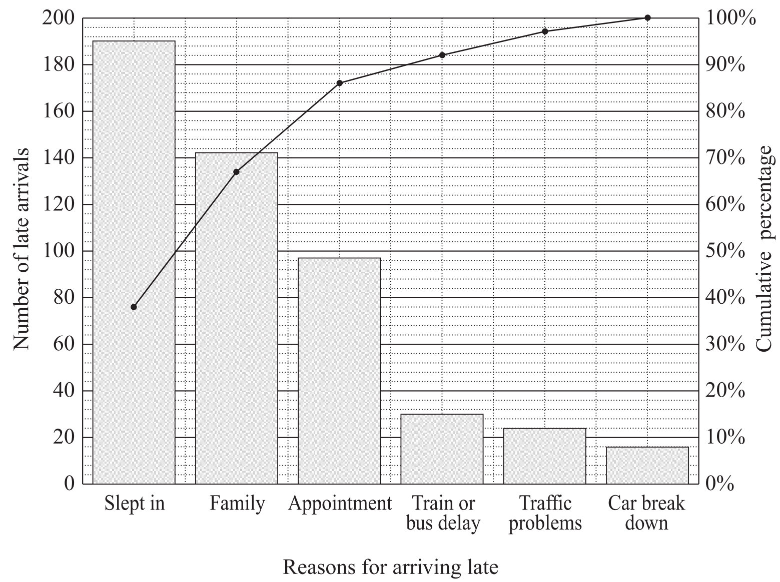
The information in the area graph is then displayed as a line graph.

Which of the following line graphs best displays the data from the area graph?



16. Statistics, STD2 S1 2019 HSC 10 MC

A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



What percentage of students gave the reason 'Train or bus delay'?

A. 6%

B. 15%

C. 30%

D. 92%

17. Statistics, STD2 S1 2009 HSC 24a

The diagram below shows a stem-and-leaf plot for 22 scores.

2	3	5	9		
3	1	4	7	9	
4	2	4	4	5	7
5	1	2	4		
6	2	3	7		
7	5	8	8	8	

i. What is the mode for this data? ***(1 mark)***

ii. What is the median for this data? ***(1 mark)***

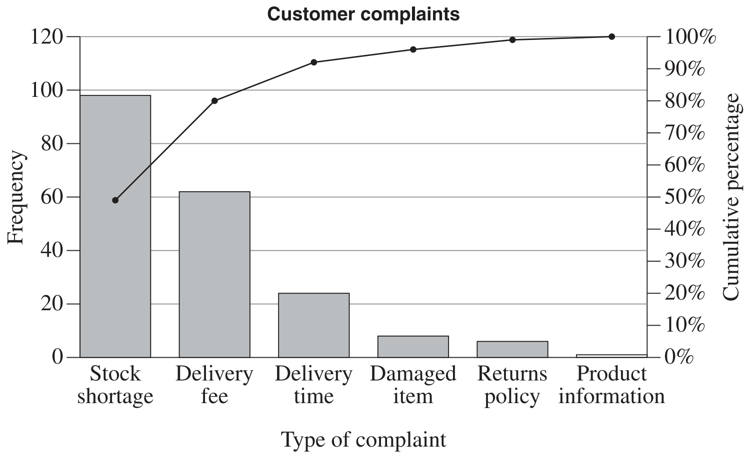
18. Statistics, 2ADV S2 2022 HSC 11

The table shows the types of customer complaints received by an online business in a month.

Type of complaint	*Frequency*	*Cumulative frequency*	*Cumulative percentage*
Stock shortage	98	98	49
Delivery fee	62	***A***	80
Delivery time	24	184	92
Damaged item	8	192	***B***
Returns policy	6	198	99
Product information	2	200	100
Total	200		

a. What are the values of A and B? ***(2 marks)***

b. The data from the table are shown in the following Pareto chart.

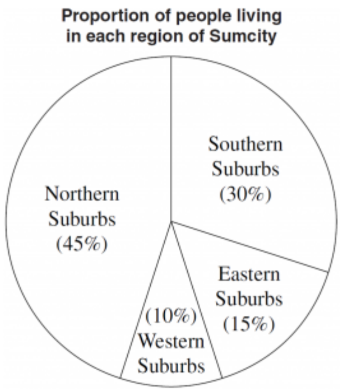


The manager will address 80% of the complaints.

Which types of complaints will the manager address? ***(1 mark)***

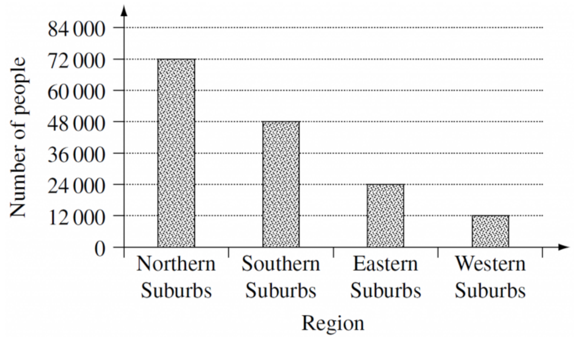
19. Statistics, STD2 S1 2005 HSC 24d

The sector graph shows the proportion of people, as a percentage, living in each region of Sumcity. There are 24 000 people living in the Eastern Suburbs.



i. Show that the total number of people living in Sumcity is 160 000. ***(1 mark)***

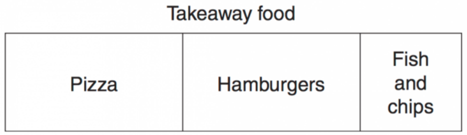
Jake used the information above to draw a column graph.



ii. The column graph height is incorrect for one region.
Identify this region and justify your answer. ***(2 marks)***

20. Statistics, STD2 S1 2008 HSC 23e

In a survey, 450 people were asked about their favourite takeaway food. The results are displayed in the bar graph.



How many people chose pizza as their favourite takeaway food? ***(2 marks)***

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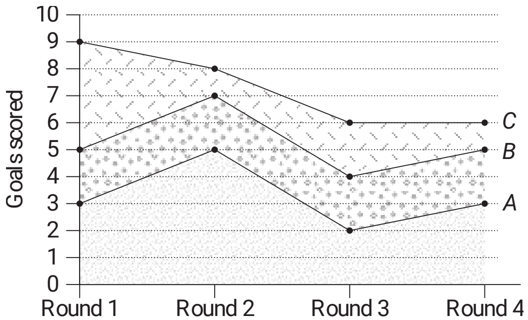
21. Statistics, STD2 S1 2017 HSC 26f

The area chart shows the number of goals scored by three hockey teams, A, B and C, in the first 4 rounds.



i. How many goals were scored by team C in round 1? ***(1 mark)***

ii. In which round did all three teams score the same number of goals? ***(1 mark)***

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22. Statistics, STD2 S1 2007 HSC 24d

Barry constructed a back-to-back stem-and-leaf plot to compare the ages of his students.

Ages of students attending Barry's Ballroom Dancing Studio

Females		*Males*
9	1	1 2 3
7	2	0 2 2 2 4 5
5	3	0 0 1 7
5 2	4	6 7
3 2 0	5	2
4 4 2 1	6	4 4

i. Write a brief statement that compares the distribution of the ages of males and females from this set of data. ***(1 mark)***

ii. What is the mode of this set of data? ***(1 mark)***

iii. Liam decided to use a grouped frequency distribution table to calculate the mean age of the students at Barry's Ballroom Dancing Studio.

For the age group 30 - 39 years, what is the value of the product of the class centre and the frequency? ***(2 marks)***

iv. Liam correctly calculated the mean from the grouped frequency distribution table to be 39.5.

Caitlyn correctly used the original data in the back-to-back stem-and-leaf plot and calculated the mean to be 38.2.

What is the reason for the difference in the two answers? ***(1 mark)***

23. Statistics, STD2 S1 2011 HSC 25d

Data was collected from 30 students on the number of text messages they had sent in the previous 24 hours. The set of data collected is displayed.

Male		*Female*
9 9 8 7 6 5 5 4 2 1	0	8 9
1 1 0 0	1	1 1 2 5 6 8 8 8
0	2	0 1 7
	3	4
	4	
	5	
	6	
1	7	

i. What is the outlier for this set of data? ***(1 mark)***

ii. What is the interquartile range of the data collected from the female students? ***(1 mark)***

24. Statistics, STD2 S1 2005 HSC 24a

i. Draw a stem-and-leaf plot for the following set of scores.

$21 \quad 45 \quad 29 \quad 27 \quad 19 \quad 35 \quad 23 \quad 58 \quad 34 \quad 27$ ***(2 marks)***

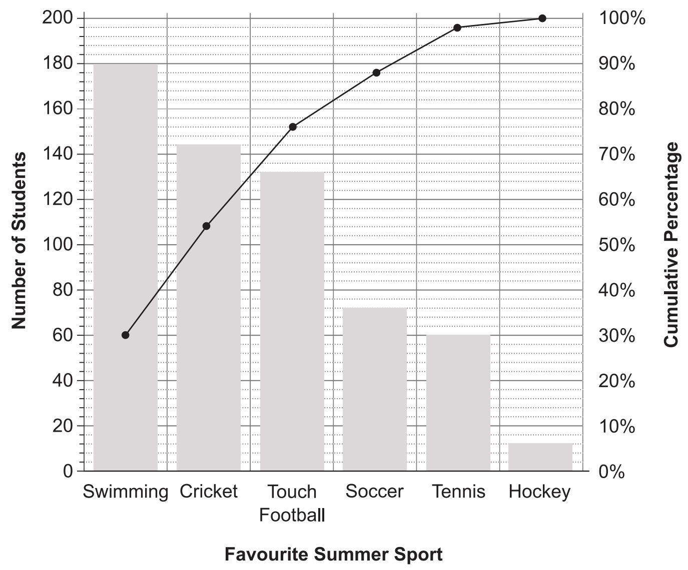
iii. What is the median of the set of scores? ***(1 mark)***

iv. Comment on the skewness of the set of scores. ***(1 mark)***

25. Statistics, STD2 S1 EQ-Bank 4

A high school conducted a survey asking students what their favourite Summer sport was.

The Pareto chart shows the data collected.



i. What percentage of students chose Hockey as their favourite Summer sport? ***(1 mark)***

ii. What percentage of students chose Touch Football as their favourite Summer sport? ***(1 mark)***

26. Statistics, STD2 S1 2013 HSC 26f

Jason travels to work by car on all five days of his working week, leaving home at 7 am each day. He compares his travel times using roads without tolls and roads with tolls over a period of 12 working weeks.

He records his travel times (in minutes) in a back-to-back stem-and-leaf plot.

Travel time (minutes)

Without tolls		*With tolls*
9	3	5 8 9 9
9 9 8 7 7 6 5 5 4 4 3 2 0	4	0 1 2 6 7 7 8 8 8 9
9 8 7 5 4 3 3 3 2 2 2 2 1 1 0	5	2 4 4 5 6 8 9
1	6	1 3 5 7
	7	0 2 8
	8	2
	9	0

i. What is the modal travel time when he uses roads without tolls? ***(1 mark)***

ii. What is the median travel time when he uses roads without tolls? ***(1 mark)***

iii. Describe how the two data sets differ in terms of the spread and skewness of their distributions. ***(2 marks)***

27. Statistics, STD2 S1 2016 HSC 29c

The ages of members of a dance class are shown in the back-to-back stem-and-leaf plot.

Women		*Men*
2	3	4 6
4 2	4	2 2 5 6 8
8 8 5 4 0 0	5	3
9 4 3 3	6	3

Pat claims that the women who attend the dance class are generally older than the men.

Is Pat correct? Justify your answer by referring to the median and skewness of the two sets of data. ***(3 marks)***

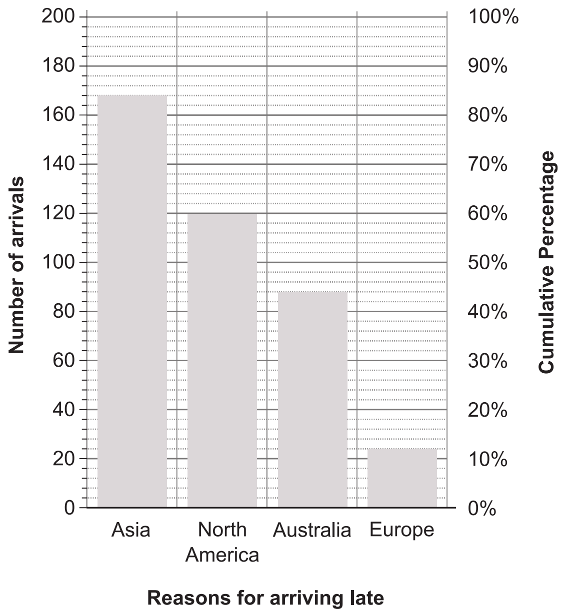
28. Statistics, STD2 S1 EQ-Bank 5

An island resort surveyed 400 guests by asking them on which continent they lived.

The table below shows the data collected.

Continent	Asia	Europe	North America	Australia
Number of guests	168	24	120	88

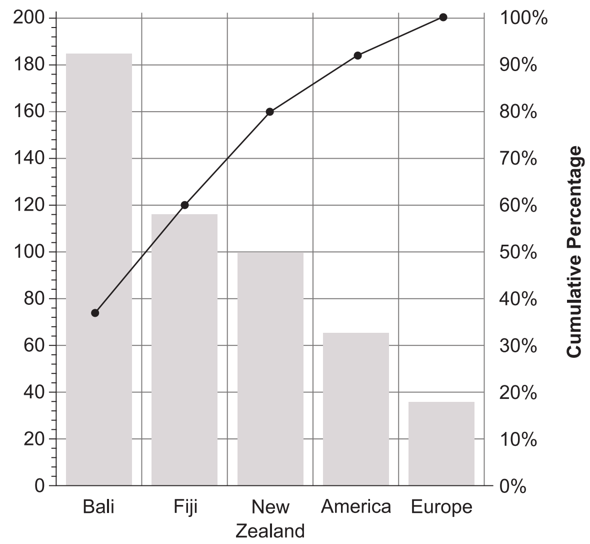
Complete the Pareto chart below to show the data collected. ***(3 marks)***



29. Statistics, STD2 S1 EQ-Bank 6

The Pareto chart below shows the data collected from a survey where people were asked to choose their favourite overseas holiday destination.

Using the chart, how many people were surveyed? ***(2 marks)***



Worked Solutions

1. Statistics, STD2 S1 2010 HSC 1 MC

$$\begin{aligned}\text{Range} &= \text{High} - \text{Low}\\ &= 9 - 0\\ &= 9\end{aligned}$$

$\Rightarrow \ C$

2. Statistics, STD2 S1 2009 HSC 2 MC

Time in carpark $=$ 3h 20m

From graph, charge will be \$18

$\Rightarrow \ C$

3. Statistics, STD2 S1 2004 HSC 8 MC

Centre angle of School X sector

$= 100^\circ$ (by measurement)

$\therefore$ Prizes won by school X

$$\begin{aligned}&= \frac{100}{360} \times 116\\ &= 32.22\ \ldots\end{aligned}$$

$\Rightarrow B$

4. Statistics, 2ADV S2 SM-Bank 5 MC

$$Q_1 = \frac{148 + 148}{2} = 148$$

$$Q_3 = \frac{158 + 160}{2} = 159$$

$$IQR = 159–148 = 11$$

$$\begin{aligned}\text{Upper fence} &= Q_3 + 1.5 \times IQR\\ &= 159 + 1.5 \times 11\\ &= 175.5\end{aligned}$$

$\Rightarrow D$

5. Statistics, STD2 S1 2012 HSC 1 MC

15 scores $\Rightarrow$ Median is 8th

$\therefore$ Median is 78

$\Rightarrow \ D$

6. Statistics, STD2 S1 2012 HSC 7 MC

$$\begin{aligned} \text{Bakery 1 Sales in June} &= \$17\ 500 \\ \text{Bakery 2 Sales in June} &= \$35\ 000 \\ \therefore\ \text{Difference in Sales} &= 35\ 000 - 17\ 500 \\ &= 17\ 500 \end{aligned}$$

$\Rightarrow\ D$

7. Statistics, STD2 S1 2006 HSC 4 MC

10 scores

$$\begin{aligned} \text{Median} &= \frac{\text{5th} + \text{6th}}{2} \\ &= \frac{28+38}{2} \\ &= 33 \end{aligned}$$

$\Rightarrow\ C$

8. Statistics, STD2 S1 2008 HSC 3 MC

$\text{Range} = \text{High}-\text{Low} = 43$

$$\begin{aligned} \therefore\ 67-\text{Low} &= 43 \\ \text{Low} &= 24 \end{aligned}$$

$\therefore\ N = 4$

$\Rightarrow\ A$

9. Statistics, STD2 S1 2018 HSC 17 MC

Area Charts are cumulative,

Consider 2015,

Cricket players $= 30$

Tennis players $= 60-30 = 30$

$\Rightarrow C$

10. Statistics, STD2 S1 2010 HSC 16 MC

♦♦ *Mean mark 35%*

26 results given in the data

$\Rightarrow$ Median is average of 13^{th} and 14^{th}

$$\begin{aligned} \therefore\ \text{Median} &= \frac{45+47}{2} \\ &= 46 \end{aligned}$$

$\Rightarrow B$

11. Statistics, STD2 S1 2006 HSC 8 MC

By elimination

Positive skew when the tail on the right side is longer.

$\therefore$ NOT B or D

A smaller standard deviation occurs when data is clustered more closely.

$\therefore$ NOT A where data is more widely spread.

$\Rightarrow\ C$

12. Statistics, STD2 S1 2018 HSC 6 MC

♦ *Mean mark 43% (a surprisingly poor result!)*

Data is skewed.

Since the "tail" is on the left had side, the data is negatively skewed.

$\Rightarrow$ C

13. Statistics, STD2 S1 SM-Bank 2 MC

Mean has increased (Y9 to Y12)

The Year 12 data is more tightly distributed around the mean.

$\therefore$ Standard deviation has decreased (Y9 to Y12)

$\Rightarrow A$

14. Statistics, STD2 S1 2015 HSC 19 MC

♦ *Mean mark 39%.*

In 1975, her life expectancy

$$\begin{aligned} &= \text{age} + \text{remaining years} \\ &= 45 + 34 \\ &= 79 \end{aligned}$$

In 1995, her life expectancy

$$\begin{aligned} &= 65 + 19.8 \\ &= 84.8 \end{aligned}$$

$$\begin{aligned} \therefore\ \text{Difference} &= 84.8 - 79 \\ &= 5.8\ \text{years} \end{aligned}$$

$\Rightarrow D$

15. Statistics, STD2 S1 2010 HSC 21 MC

♦♦♦ *Mean mark 24%*
COMMENT: Area graph questions have proven very challenging over recent years. Review this area.

Since the area in the profit graph is constant and then gradually decreases after a point

$\Rightarrow B$

16. Statistics, STD2 S1 2019 HSC 10 MC

Train or bus delay (%)

$$\begin{aligned} &= 92\text{–}86 \\ &= 6\% \\ &\Rightarrow A \end{aligned}$$

♦♦♦ *Mean mark 18%.*

17. Statistics, STD2 S1 2009 HSC 24a

i. Mode $= 78$

ii. 22 scores

$\Rightarrow$ Median is the average of 11th and 12th scores

$$\begin{aligned} \therefore \text{Median} &= \frac{45+47}{2} \\ &= 46 \end{aligned}$$

18. Statistics, 2ADV S2 2022 HSC 11

a. $A = 98 + 62 = 160$

$$\% \text{ Damaged items } = \frac{8}{200} \times 100 = 4\%$$

Cumulative % after damaged items $= 96\%$

$B = 92 + 4 = 96$

b. The right hand side cumulative frequency percentage shows that 80% of all complaints received concern stock shortages and delivery fees.

$\therefore$ The manager will address stock shortages and delivery fees.

19. Statistics, STD2 S1 2005 HSC 24d

i. Let the population of Sumcity $= P$

$$\begin{aligned} 15\% \times P &= 24\ 000 \\ \therefore P &= \frac{24\ 000}{0.15} \\ &= 160\ 000 \ldots \text{ as required} \end{aligned}$$

ii. Western Suburbs population

$$\begin{aligned} &= 10\% \times 160\ 000 \\ &= 16\ 000 \end{aligned}$$

The column graph has this population as 12 000 people which is incorrect.

20. Statistics, STD2 S1 2008 HSC 23e

Number of people who chose pizza

$$\begin{aligned} &= \frac{\text{Length of pizza section}}{\text{Total length of bar}} \times 450 \\ &\approx \frac{7}{18} \times 450 \\ &\approx 175 \end{aligned}$$

$\therefore$ 175 people chose pizza.

COMMENT: This question required measurement of the actual image on the exam. The same methodology works here.

21. Statistics, STD2 S1 2017 HSC 26f

i. Goals by team C $= 9\text{–}5$

$= 4$

ii. In Round 3 (all teams scored 2 goals).

22. Statistics, STD2 S1 2007 HSC 24d

i. More males attend than females and a higher proportion of those are younger males, with the distribution being positively skewed. Female attendees are generally older and have a negatively skewed distribution.

ii. Mode $= 64$ (4 times)

iii.

$$\begin{aligned} \text{Class centre} &= \frac{30+39}{2} \\ &= 34.5 \end{aligned}$$

Frequency $= 5$

$$\begin{aligned} \therefore\ &\text{Class centre} \times \text{frequency} \\ &= 34.5 \times 5 \\ &= 172.5 \end{aligned}$$

iv. The difference in the answers is due to the class centres used in group frequency tables distorting the mean value from the exact data.

23. Statistics, STD2 S1 2011 HSC 25d

i. Outlier is 71

ii. Lower quartile $= 9$ (4th female data point)

Upper quartile $= 20$ (11th female data point)

$\therefore$ Interquartile range (female) $= 20 - 11 = 9$

♦♦ *Mean mark 34%*
COMMENT: Ensure you can quickly and accurately find quartile values using stem and leaf graphs!

24. Statistics, STD2 S1 2005 HSC 24a

i.

Stem	Leaf
1	9
2	1 3 7 7 9
3	4 5
4	5
5	8

ii. 10 scores

$$\therefore \text{Median} = \frac{(5\text{th} + 6\text{th})}{2} = \frac{27 + 29}{2} = 28$$

iii. The data has a tail that stretches to the right

$\therefore$ Data is positively skewed.

25. Statistics, STD2 S1 EQ-Bank 4

i. Method 1

Percentage who chose hockey

$= $ cum freq (at hockey column) – cum freq (at tennis column)
$= 100 - 98$
$= 2\%$

Method 2

Column 1 $\rightarrow$ 180 people $= 30\%$

Total surveyed $= 180 \div 0.3 = 600$

$$\therefore \text{Hockey } \% = \frac{12}{600} \times 100 = 2\%$$

ii. Percentage who chose touch football

$= $ cum freq (at touch football column) – cum freq (at cricket column)
$= 76 - 54$
$= 22\%$

26. Statistics, STD2 S1 2013 HSC 26f

i. Modal time $= 52$ minutes

♦ *Mean mark 36%*
MARKER'S COMMENT: Finding a median proved challenging for many students. Take note!

ii. 30 times with no tolls

Median $=$ Average of 15th and 16th

$$= \frac{50 + 51}{2} = 50.5 \text{ minutes}$$

♦ *Mean mark 39%*

iii. Spread

Times without tolls have a much tighter spread (range $= 22$) than times with tolls (range $= 55$).

Skewness

Times without tolls shows virtually no skewness while times with tolls are positively skewed.

27. Statistics, STD2 S1 2016 HSC 29c

Women:
The median is 55 in a data set that is negatively skewed.
Men:
The median is 45 in a data set that is positively skewed.
$\therefore$ Pat is correct.

♦ *Mean mark 44%.*

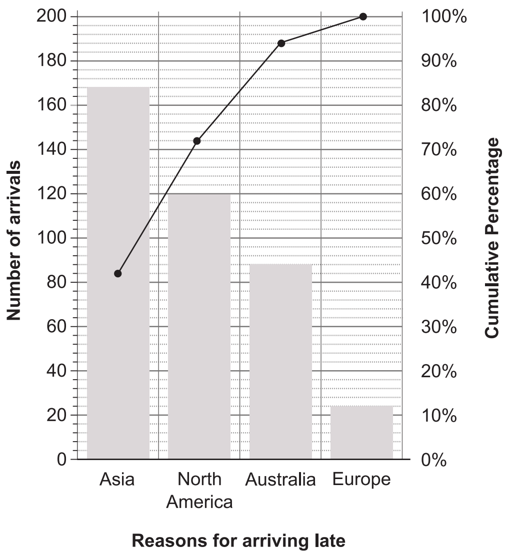
28. Statistics, STD2 S1 EQ-Bank 5

$$\%\ \text{Asia} = \frac{168}{400} = 42\%$$

$$\%\ \text{North America} = \frac{120}{400} = 30\%$$

$$\%\ \text{Australia} = \frac{88}{400} = 22\%$$

$$\%\ \text{Europe} = \frac{24}{400} = 6\%$$



29. Statistics, STD2 S1 EQ-Bank 6

Percentage of people whose chose NZ
$= 80–60$
$= 20\%$

Number of people who chose NZ
$= 100$

$20\% \times \text{Total surveyed} = 100$
$\therefore \text{Total surveyed} = 500$